MAT215: Complex Variables And Laplace Transformations

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LECTURE-01

Complex plane

As a set, the complex number field $\mathbb{C}$ be the set $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$. The set is a plane, so we call it the complex plane. To make it a field, we define addition and product:

$$(a,b)+(c,d) \stackrel{\text{def}}{=} (a+c, b+d)$$

$$(a,b)(c,d) \stackrel{\text{def}}{=} (ac-bd, bc+da)$$

Why do we define the operations in such a way? One answer is to mimic the complex operation. Another is to consider the multiplication as the map $z_1 \mapsto z_1 z_2$ (which is a real-linear operator).

$$a+ib = a\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + b\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

Problem

Show that the additive identity is $\mathbf{0} = (0,0)$*, the multiplicative identity is* $\mathbf{1} = (1,0)$ *and the multiplicative inverse* $(a,b)^{-1} = \left(\frac{a}{a^2+b^2}, \frac{-b}{a^2+b^2}\right)$

Complex numbers

Problem

Find the real numbers x and y such that $3x+2iy-ix+5y=7+5i$

Problem

Prove that every real-linear operator on $\mathbb{C}$, that is every 2×2 real matrix M, can be represented by two complex numbers ξ and ζ and the formula $z\mapsto \xi z+\zeta\bar{z}$.

Hint: *Use $T(z)=T(x+iy)$*

Complex number

Given a complex number $z \stackrel{\text{def}}{=} x + iy$, its "evil twin" is the complex conjugate of z:

$$\bar{z} \stackrel{\text{def}}{=} x - iy$$

Find the real and imaginary parts.
We can show that,

$$x^3 + y^3 + 3ixy = \left(\frac{z+\bar{z}}{2}\right)^3 + \left(\frac{z-\bar{z}}{2i}\right)^3 + 3i\left(\frac{z+\bar{z}}{2}\right)\left(\frac{z-\bar{z}}{2i}\right)$$

Problem

Guess all the properties of Conjugate.

z and $\bar{z}$ are not independent variables, why?

Problem

Problem

Find the square root of $3+4i$.

Problem

Solve the equation $z^2+(1+i)z+\left(-\frac{3}{4}-\frac{1}{2}i\right)=0$.

Argument

The function $\mathbf{Arg}(z) : \mathbb{C}\backslash\{0\} \to (-\pi, \pi]$ is defined as follows:

$$\mathrm{Arg}(z) = ??$$

Thus, if $z = r(\cos\Theta + i\sin\Theta)$, with $r > 0$ and $-\pi < \Theta \leq \pi$, then

$$\arg(z) = \{\mathrm{Arg}(z) + 2n\pi \mid n \in \mathbb{Z}\}.$$

Polar form

Properties of Modulus:

$$|z_1 \cdot z_2| = |z_1| \cdot |z_2|$$

$$\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$$

$$|z^m| = |z|^m$$

Properties of Arguments:

$$\mathrm{Arg}(z_1 \cdot z_2) = \mathrm{Arg}(z_1) + \mathrm{Arg}(z_2)$$

$$\mathrm{Arg}\left(\frac{z_1}{z_2}\right) = \mathrm{Arg}(z_1) - \mathrm{Arg}(z_2)$$

$$\mathrm{Arg}(z^m) = m\,\mathrm{Arg}(z)$$

https://scipp.ucsc.edu/~haber/ph116A/arg_11.pdf

Weird properties

$$\left((-1)^{\frac{1}{2}}\right)^2 = i^2 = -1 \neq 1 = \left((-1)^2\right)^{\frac{1}{2}}$$

$$\sqrt{(-4)(-4)} = \sqrt{-4}\sqrt{-4} = 2i \cdot 2i = -4 \neq 4 = \sqrt{16}$$

For real numbers ($a > 0$),

$$a^z := e^{z \ln a}$$

where ($\ln a$) is a single real number. So:

$$(a^m)^n = e^{n \ln(a^m)} = e^{nm \ln a} = a^{mn}.$$

Everything works because ln is single-valued.

Polar form

Express each of the following complex numbers in polar form:

(a) $3+4i$

(b) $1+1i$

problems

Problem

Solve the equation: $e^{4z} = i$

Problem

Solve for x *and* y*,*

$$\left(\frac{3}{2} + \frac{\sqrt{3}}{2}i\right)^{2024} = 3^{1012}(x + iy)$$